CHAPTER 3 · DERIVATIONS

DYNAMICS

2
derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FORCE AS RATE OF CHANGE OF MOMENTUM

SYLLABUS 3.1(C)

RESULT $F = m\frac{dv}{dt} + v\frac{dm}{dt}$

1. Newton's second law: the resultant force equals the rate of change of momentum, $F = \frac{dp}{dt}$.
2. Momentum $p = mv$, so $F = \frac{d(mv)}{dt}$.
3. Product rule: $\mathbf{F = m\frac{dv}{dt} + v\frac{dm}{dt}}$.
4. Constant mass: $F = m\frac{dv}{dt} = ma$. Constant velocity (mass changing, e.g. sand falling onto a moving belt): $F = v\frac{dm}{dt}$.

Use only the cases of constant m or constant v.

2 VERIFY CONSERVATION OF LINEAR MOMENTUM

SYLLABUS 3.2(B)

RESULT $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

1. Bodies A and B collide. During the contact time Δt, B pushes A with F_{BA} and A pushes B with F_{AB}.
2. Newton's third law: $F_{BA} = -F_{AB}$.
3. Newton's second law: $F_{BA} = \frac{\Delta p_A}{\Delta t}$ and $F_{AB} = \frac{\Delta p_B}{\Delta t}$.
4. So $\frac{\Delta p_A}{\Delta t} = -\frac{\Delta p_B}{\Delta t}$, giving $\Delta p_A + \Delta p_B = 0$.
5. $m_A(v_A - u_A) + m_B(v_B - u_B) = 0$, so $\mathbf{m_Au_A + m_Bu_B = m_Av_A + m_Bv_B}$: total momentum is conserved when no external force acts.

